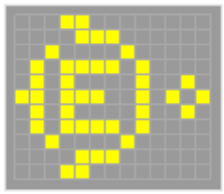
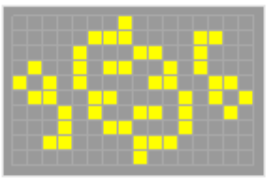


pseudo-barberpole



B-heptomimo



babbling brook

Homework #2 (due Oct. 30)

- Read up a bit on Ising models. Besides the usual (Wikipedia, etc.) a few good starting points are:

  https://people.mines.edu/kbachman/ising/

  https://peteryoung.sites.ucsc.edu/research/ising-model-early-visualizations/

  https://arxiv.org/abs/0803.0217

- Idem for Conway's game of life, Turing machines, cellular automata, collective (swarm) behavior, etc. See links posted in Canvas. Lots to ponder about.

On your own (no help from Claude, other than for debugging):

- Further develop the posted code "life.f" to include the biological rules described there. Make sure your code works, by setting off a "glider" (coordinates (33,33), (33,34), (33,35), (34,33), (35,34)), or any other beast from the menagerie described in the links posted. Remember, the one precaution to keep in mind is not accessing elements of the game matrix outside of bounds, when tallying the number of live next neighbors, as this will lead to a runtime error (segmentation fault).

Vibe-coding your way into extinction:

- Any conclusions from repeating the game (while changing random number generator seed) and keeping some tallies? (e.g., long-term stability as a function of initial filling factor, minimum initial colony size for a certain longevity, longevity vs. size of Petri dish for a fixed filling factor, prob. distribution of game duration until stagnancy, etc. Anything you deem of sufficient interest to study.)

- Let your hair down and mess with the rules of life. *Suggestions*: longer range interactions (e.g., swarm behavior), collaborative or war games (tribes!), introduction of an element of randomness (i.e., deviation from a purely deterministic game), use of a toric board, so on, and so forth. The whole beauty is that a set of rules that "make sense" may actually result into the **emergence** of surprising collective/coherent behaviors.

  Extension (only for uber-geeks): beat me at "peg solitaire" (see shallowblue.f code). Develop a more successful algorithm than mine.